Experiment No. 6

Insurance Claim Automation Suite: Workflow Options in Zoho Creator

1. Aim and Objective

To design and build an automation layer on the existing **Insurance Claim** application in Zoho Creator, using each of the eight workflow options: **Form Workflows, Schedules, Approvals, Payments, Blueprints, Batch Workflows, Report Workflows and Functions**. The aim is for a claim to move from submission to settlement with minimal manual intervention.

Before automation, every downstream action depended on a person: acknowledging claims, chasing claimants for missing documents, escalating high-value claims for senior approval, initiating the settlement payout, processing claims in bulk and recalculating the net payable amount. The suite below replaces those manual steps with event-driven and time-driven rules and keeps an auditable trail.

2. Workflow Options and Mapping

Workflow option	Workflow name	Trigger / frequency	Purpose
Form Workflow	Claim Submission Workflow	On Create (successful form submission)	Acknowledgement email with claim reference to the customer; notify claims team
Schedule	Daily_Missing_Document_Reminder_Schedule	Daily at 10:00 AM	Remind claimants whose supporting documents are missing
Approval	HighValue_Claim_Approval	On review of the claim	Senior approval of high-value claims before settlement
Payment	Claim_Settlement_Payout_Workflow	On collecting payment	Settlement payout once the claim is Approved; updates settlement status
Blueprint	Claim_Lifecycle_Blueprint	Every record of the Insurance Claim form	Controlled stages with role-based transitions
Batch Workflow	Batch_Claim_Archival	Scheduled run at 10:00 AM	Process claims in controlled batches for archival
Report Workflow	OnEdit_Recalculate_Settlement_Workflow	On Edit in the claims report	Recalculate the net payable amount when figures are edited in the report
Function	calculate_settlement_payout	Called from workflows / buttons	Reusable Deluge function that computes the settlement payout

3. Application Setup

The application **Insurance Claim1** was created in Zoho Creator under the account of the Super Admin user. It contains three forms: **Customer Details**, **Insurance Policy** and **Insurance Claim**. The Insurance Claim form holds the data that every workflow depends on. Figure 1 shows the form in the live application, with the fields that feed the automations: Customer, Policy Number, Customer Email (used for every notification), Claim Date, Claim Type, Claim Amount, Approved Amount (used by the approval and payout), Description, Supporting Documents (checked by the reminder schedule) and Claim Status.

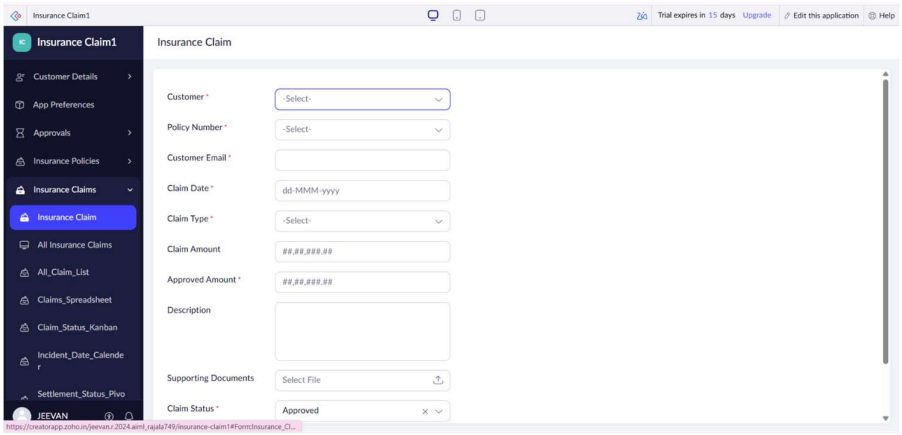


Figure 1: Insurance Claim form in the live Insurance Claim1 application

The sidebar of the application lists the supporting reports and views: All Insurance Claims, All_Claim_List, Claims_Spreadsheet, Claim_Status_Kanban, Incident_Date_Calender and Settlement_Status_Pivot, plus an Approvals section and the Customer Details and Insurance Policies sections.

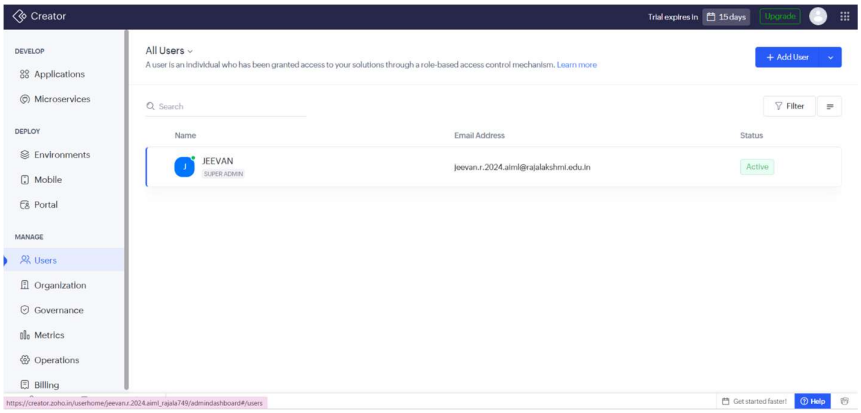


Figure 2: Users page in Zoho Creator showing the application Super Admin (role-based access control)

Role-based access is the basis for the Blueprint transition permissions described in Section 4.5.

4. Workflow Implementation

4.1 Form Workflow: Claim Submission Workflow

Event: Created, on successful form submission of Insurance Claim. **Action:** send an acknowledgement email to the customer with the claim details and notify the claims team of the new submission.

- Configured to fail gracefully: if the customer email is invalid, an error is logged and the submission is not blocked.
- Uses a *Send Mail* action inside an *if* check on the email field.

Sample Deluge logic:

```
if(input.Customer_Email != null && input.Customer_Email.contains("@"))
{
    sendmail
    [
        from: zoho.adminuserid
        to: input.Customer_Email
        subject: "Claim " + input.ID + " received"
        message: "Dear Customer, your claim reference ID is " + input.ID + "."
    ]
}
else
{
    info "Invalid customer email for claim " + input.ID;
}
```

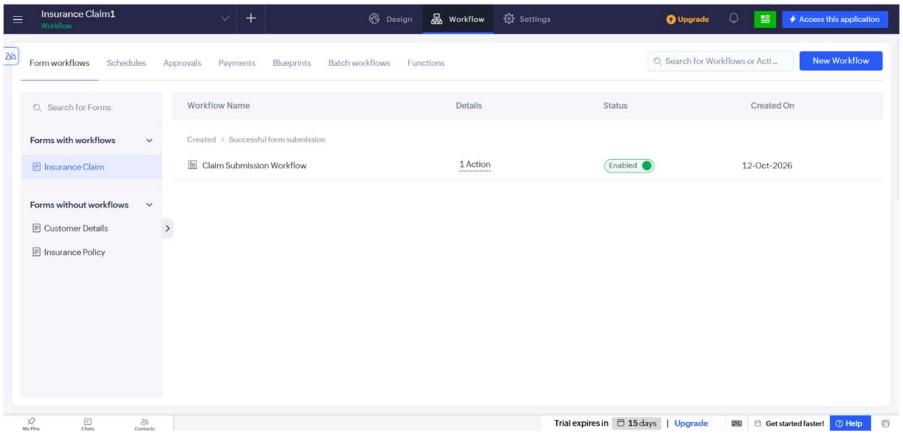


Figure 3: Workflow tab of the Insurance Claim1 app. Claim Submission Workflow (Created > Successful form submission) is enabled for the Insurance Claim form with 1 action.

4.2 Schedule: Daily_Missing_Document_Reminder_Schedule

A time-driven custom schedule that runs every day at 10:00 AM. It fetches the claims that are not yet marked document-complete and have no supporting documents, and sends each claimant a reminder listing the documents required (repair estimate, FIR copy, etc.). Reminders stop automatically once the claim is marked document-complete, and are capped at 5 per claim before escalating to a manual follow-up call.

```
pending = Insurance_Claim[Supporting_Documents == null && Claim_Status != "Rejected"];
for each c in pending
{
    sendmail [ from: zoho.adminuserid  to: c.Customer Email
               subject: "Reminder: supporting documents pending"
               message: "Please upload the required supporting documents for claim " + c.ID +
"." ]
}
```

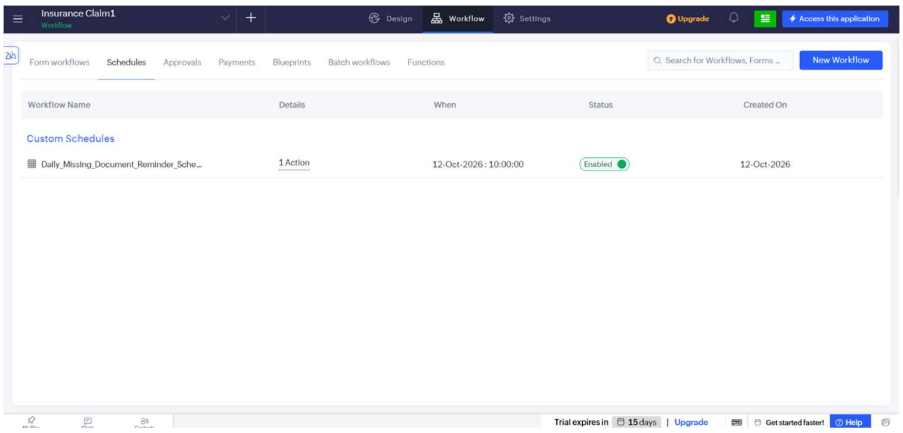


Figure 4: Schedules tab showing Daily_Missing_Document_Reminder_Schedule (1 action, daily at 10:00:00, Enabled)

4.3 Approval: HighValue_Claim_Approval

The approval runs on review of a claim of the Insurance Claim form. Only claims whose value exceeds the configured high-value threshold are routed for approval; smaller claims are settled directly by the claims team. The Senior Claims Manager is the approver, with a fallback approver if the primary approver is unavailable. The record stays at *Pending Senior Approval* until the approver approves or rejects it from the approval inbox. The workflow has 2 actions (one on approval, one on rejection).

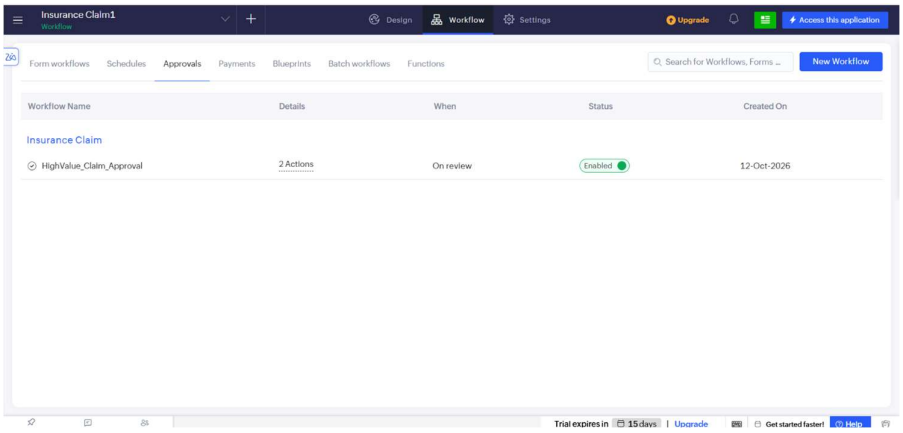


Figure 5: Approvals tab showing HighValue_Claim_Approval (2 actions, On review, Enabled)

4.4 Payments: Claim_Settlement_Payout_Workflow

Triggered when a claim is approved. It uses the payment gateway / bank-transfer connector to initiate the settlement payout based on the approved amount. On success, Settlement Status becomes **Paid** and the transaction reference is stored; on failure the record is flagged for finance follow-up. The status-check callback is retried up to 3 times before the transaction is flagged as failed and finance is alerted. The workflow has 2 actions (success and failure). The page shows a banner that a payment gateway must be configured before live online transactions can be processed.

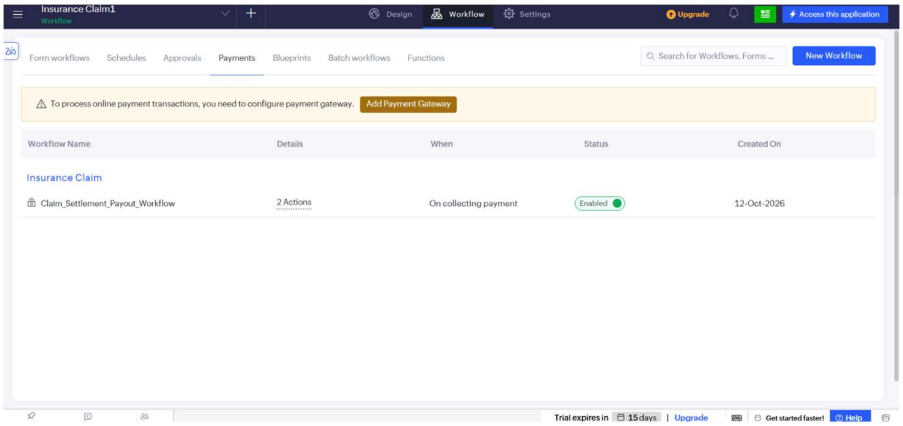


Figure 6: Payments tab showing Claim_Settlement_Payout_Workflow (2 actions, On collecting payment, Enabled)

4.5 Blueprint: Claim_Lifecycle_Blueprint

The blueprint is created from the Insurance Claim form. The Blueprint builder screen (Figure 7) shows the canvas with the stages and transitions. The lifecycle is:

Submitted → Document Verification → Investigation → Approved → Settled



Figure 7: Claim_Lifecycle_Blueprint in the Blueprint builder (stages: Submitted, Document Verification, Investigation, Approved, Settled; transitions: Verify Documents, Investigate, Approve Claim, Settle Payment)

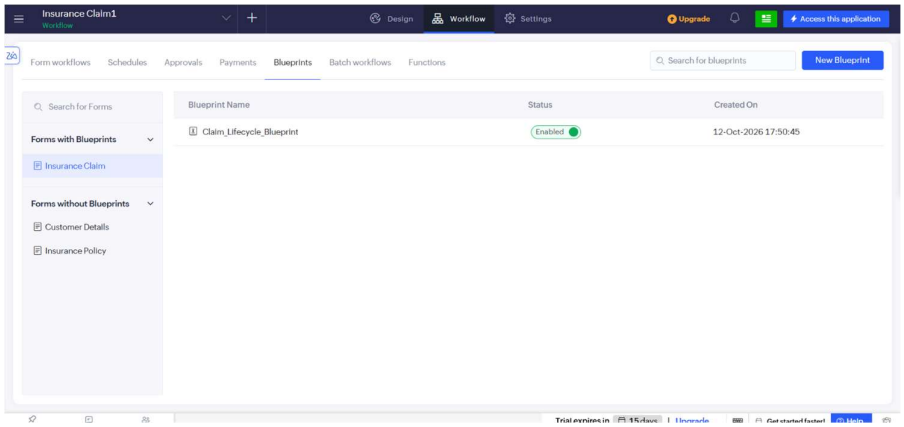


Figure 8: Blueprints tab showing Claim_Lifecycle_Blueprint enabled for the Insurance Claim form (created 12-Oct-2026 17:50:45)

Transition	Allowed role	Condition
Submitted → Document Verification (Verify Documents)	Claims Processing Team	Comment mandatory
Document Verification → Investigation (Investigate)	Claims Processing Team	Comment mandatory; documents marked complete
Investigation → Approved (Approve Claim)	Senior Claims Manager	Comment mandatory; approval granted if high-value
Approved → Settled (Settle Payment)	Finance / Claims Manager	Settlement Status = Paid; comment mandatory

Every transition requires a comment, so the Blueprint view gives management an auditable trail of who moved each claim and when. A front-desk user cannot mark a claim Settled without the intermediate stages being completed.

4.6 Batch Workflow: Batch_Claim_Archival

A batch workflow on the Insurance Claim form, enabled and triggered on schedule (first run 12-Oct-2026 at 10:00:00). It selects the claims that have reached their final stage and processes them in batches rather than all at once, so that a large claims backlog does not time out the workflow engine. The batch size is configurable.

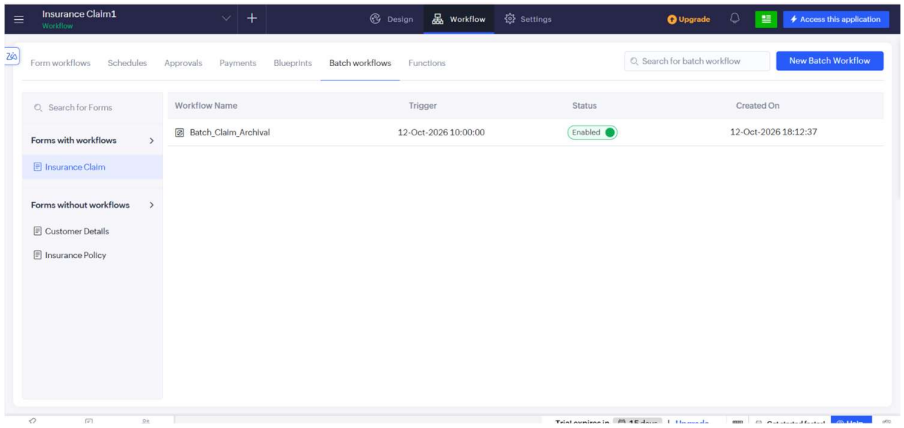


Figure 9: Batch workflows tab showing Batch_Claim_Archival (trigger 12-Oct-2026 10:00:00, Enabled)

4.7 Report Workflow: OnEdit_Recalculate_Settlement_Workflow

Fires on edit of a record in the claims report and only when **Claim Amount** or **Approved Amount** is changed from the report grid, so unrelated edits do not trigger recalculation. It keeps the payable figure consistent without requiring a full form resubmission.

```
if(input.Claim_Amount != null && input.Approved_Amount != null)
{
    // payable never exceeds the claimed amount
    input.Approved_Amount = min(input.Approved_Amount, input.Claim_Amount);
}
```

4.8 Function: calculate_settlement_payout

A reusable Deluge function in the Default namespace (return type *void*) that validates the record, computes the settlement payout from the approved amount, and updates the claim. It is written once and can be called from a form workflow, a button or a schedule. Input validation makes it fail safely when the Approved Amount is empty.

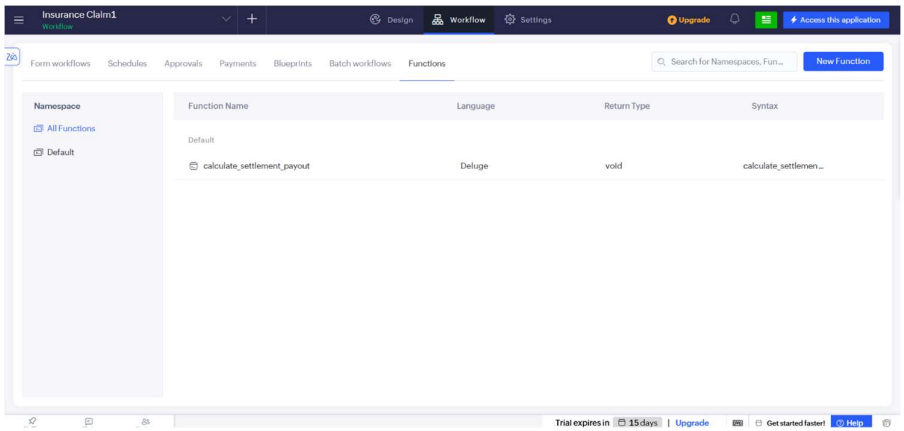


Figure 10: Functions tab showing the calculate_settlement_payout function (Deluge, return type void)

```
void calculate_settlement_payout(int recordID)
{
    rec = Insurance_Claim[ID == recordID];
    if(rec.Approved_Amount == null)
    {
        info "Payout skipped: approved amount missing for claim " + recordID;
        return;
    }
    rec.Approved_Amount = rec.Approved_Amount;
    sendmail
    [
        from: zoho.adminuserid
        to: rec.Customer_Email
        subject: "Settlement for claim " + rec.ID
        message: "Your approved settlement amount is " + rec.Approved_Amount
    ]
}
```

The function is called with thisapp.calculate_settlement_payout(input.ID); from the form workflow that fires when Claim Status is set to Approved, and from a custom button in the report.

5. End-to-End Flow

Step	What happens	Workflow option
1	Claimant submits the Insurance Claim form; acknowledgement is sent	Form Workflow
2	Each morning, claims with missing documents trigger a reminder	Schedule
3	Record moves through the lifecycle stages, each with a comment	Blueprint
4	High-value claims are routed to the senior manager for approval	Approval
5	Manual edits in the report recompute the payable amount	Report Workflow
6	Once approved, the settlement payout is initiated and Settlement Status updated	Payments
7	The settlement amount is computed and communicated through the reusable function	Function
8	Final-stage claims are processed in batches for archival	Batch Workflow

6. Configuration and Governance Rules

Workflow	Rule applied
Form Workflow	Fail gracefully and log an error on invalid customer email; never block submission
Schedule	Maximum of 5 reminders per claim before escalating to a manual follow-up call
Approval	Single approver role (Senior Claims Manager) with a fallback approver
Payments	Retry gateway status callback up to 3 times before flagging failure and alerting finance
Blueprint	Mandatory comment on every stage transition
Batch Workflow	Batch size configurable; records processed in groups
Report Workflow	Fires only when Claim Amount or Approved Amount changes
Function	Input validation on Approved Amount before processing

7. Learning Outcomes

- Event-driven automation (Form and Report Workflows) reacts to a data change, while a Schedule runs on the clock.
- Blueprints give a controlled, auditable process, and Approvals add a human checkpoint inside it.
- Batch Workflows split large jobs into groups so the engine does not time out.
- A Function is written once and reused from workflows, buttons and schedules.
- Payments workflows can handle settlement payouts, not just fee collection.

8. Evidence Included

Workflow Builder configuration screens: Form Workflows (Fig. 3), Schedules (Fig. 4), Approvals (Fig. 5), Payments (Fig. 6), Batch Workflows (Fig. 9) and Functions (Fig. 10). Blueprint visual diagram with stages and transitions (Fig. 7, 8). Application form and role-based access (Fig. 1, 2). Deluge scripts are given in Sections 4.1, 4.2, 4.7 and 4.8.

9. Conclusion

The eight workflow options cover the full claims lifecycle: instant acknowledgement, scheduled document reminders, controlled stage movement, senior approval for high-value claims, accurate settlement figures, automated payouts, reusable settlement logic and batch processing at scale. Together they reduce manual effort and make the process consistent and auditable.

Result:

Thus, all eight workflow options were successfully implemented in the **Insurance Claim1** application in Zoho Creator: Claim Submission Workflow (Form Workflow), Daily_Missing_Document_Reminder_Schedule (Schedule), HighValue_Claim_Approval (Approval), Claim_Settlement_Payout_Workflow (Payments), Claim_Lifecycle_Blueprint (Blueprint), Batch_Claim_Archival (Batch Workflow), OnEdit_Recalculate_Settlement_Workflow (Report Workflow) and calculate_settlement_payout (Function). All configured workflows show the status **Enabled** in the Workflow Builder, and the claims lifecycle now runs from submission to settlement with minimal manual intervention.